Milestone 03

Answer Key

Environment

Update the YAML with your first and last name.

Load the packages.

```
library(tidyverse)
library(haven)
library(gssr)
library(gssrdoc)
library(summarytools)
library(Rmisc) # load the new package
library(gtsummary) # load the new package
```

Load the 2022 GSS data

```
# ADD YOUR CODE HERE TO LOAD THE GSS 2022 DATA
gss22 <- gss_get_yr(2022)
```

```
Fetching: https://gss.norc.org/documents/stata/2022_stata.zip
```

You'll be working with the following GSS variables:

- `meovrwrk`: Men hurt family when focus on work too much
- `sex`: Respondent's gender
- `lifenow` R's rating of life overall now from 0-10
- `educ` Highest year of school completed

Code, Output, Meaning

Use R to complete the checkpoints below. Show your work (e.g., R code chunks) where appropriate. Add narrative (text outside code chunks) or comments (text inside the code chunks) throughout.

Show your df and variable changes as you go (e.g., `table()`, `head()`). Reference specific statistics (where appropriate) from your output to justify your answers. Explain what the values tell you about the data; interpret their meaning in relation to the question.

Checkpoint 01: Create a new dataframe

A. Create a df named `my_data` with `meovrwrk`, `sex`, `marital`, `lifenow`, and `educ`. Then, use `drop_na()` to remove rows with missing data.

```
my_data <- gss22 |>
  select(meovrwrk, sex, marital, lifenow, educ) |>
  drop_na()

head(my_data)
```

```
# A tibble: 6 x 5
  meovrwrk                        sex        marital           lifenow   educ
  <dbl+lbl>                       <dbl+lbl>  <dbl+lbl>         <dbl+lbl> <dbl+lb>
1 2 [agree]                       2 [female] 5 [never married]  8        16 [4 y~
2 3 [neither agree nor disagree] 2 [female] 5 [never married] 10 [10 -~ 12 [12t~
3 4 [disagree]                    2 [female] 1 [married]        8        12 [12t~
4 3 [neither agree nor disagree] 2 [female] 1 [married]        8        16 [4 y~
5 3 [neither agree nor disagree] 2 [female] 1 [married]        8        14 [2 y~
6 2 [agree]                       1 [male]   5 [never married]  9        13 [1 y~
```

Checkpoint 02: Clean and create variables

A. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `hurt_num` that groups respondents as follows:

- 1 if `meovrwrk` is ‘strongly agree’ or ‘agree’
- 0 if `meovrwrk` is equal to anything else

```r
my_data <- my_data |>
  mutate(
    hurt_num = case_when(
      meovrwrk %in% 1:2 ~ 1,
      TRUE ~ 0
    ))

table(my_data$meovrwrk, my_data$hurt_num) # check the recode
```

```
     0   1
  1   0 123
  2   0 535
  3 408   0
  4 308   0
  5  74   0
```

```r
table(my_data$hurt_num) # view counts
```

```
  0   1
790 658
```

B. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `college` that groups respondents as follows:

- "No college degree" if `educ` is less than 16
- "College degree" if `educ` 16 or more

```r
my_data <- my_data |>
  mutate(
    college = case_when(
      educ < 16 ~ "No college degree",
      educ >= 16 ~ "College degree"
  ))

table(my_data$educ, my_data$college)
```

```
     College degree No college degree
```

```
  2               0               1
  3               0               1
  4               0               1
  5               0               3
  6               0               5
  7               0               3
  8               0              13
  9               0              10
  10              0              12
  11              0              36
  12              0             334
  13              0             108
  14              0             193
  15              0              90
  16            350               0
  17             74               0
  18            100               0
  19             37               0
  20             77               0
```

```r
table(my_data$college)
```

```

   College degree No college degree
              638               810
```

C. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `edu_category` that groups respondents as follows:

- "High school or less" if `educ` is 12 or less
- "Some college" if `educ` is equal to or greater than 13 and equal to or less than 15
- "College or more" if `educ` is 16 or more

BONUS: Then, use `factor()` to put the categories into a meaningful order

```r
my_data <- my_data |>
  mutate(
    edu_category = case_when(
      educ <= 12 ~ "High school or less",
      educ >= 13 & educ <= 15 ~ "Some college",
      educ >= 16  ~ "College or more"
  ))
```

```r
# Convert edu_category to an ordered factor
my_data <- my_data |>
  mutate(
    edu_category = factor(
      edu_category,
      levels = c("High school or less", "Some college", "College or more")
      ))

table(my_data$educ, my_data$edu_category)
```

```
    High school or less Some college College or more
  2                   1            0               0
  3                   1            0               0
  4                   1            0               0
  5                   3            0               0
  6                   5            0               0
  7                   3            0               0
  8                  13            0               0
  9                  10            0               0
  10                 12            0               0
  11                 36            0               0
  12                334            0               0
  13                  0          108               0
  14                  0          193               0
  15                  0           90               0
  16                  0            0             350
  17                  0            0              74
  18                  0            0             100
  19                  0            0              37
  20                  0            0              77
```

```r
table(my_data$edu_category)
```

```
High school or less        Some college     College or more
                419                 391                 638
```

D. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `marstat` that groups respondents as follows:

- "Married" if `marital` is 'married'
- "Single" if `marital` is 'never married'
- "Prev married" if `marital` is 'widowed', 'divorced', or 'separated'

BONUS: Then, use `factor()` to put the categories into a meaningful order

```
my_data <- my_data |>
  mutate(
    marstat = case_when(
      marital == 1 ~ "Married",
      marital == 5 ~ "Single",
      marital %in% 2:4    ~ "Prev married"
  ))

# Convert marstat to an ordered factor
my_data <- my_data |>
  mutate(
    marstat = factor(
      marstat,
      levels = c("Single", "Married", "Prev married")
      ))

table(my_data$marital, my_data$marstat)
```

```

    Single Married Prev married
  1      0     631            0
  2      0       0           22
  3      0       0          225
  4      0       0           39
  5    531       0            0
```

```
table(my_data$marstat)
```

```

      Single      Married Prev married
         531          631          286
```

E. Use `zap_missing()`, `as_factor()`, and `droplevels()` to make `sex` and `meovrwkr` into clean factor variables.

```r
my_data$sex <- zap_missing(my_data$sex)
my_data$sex <- as_factor(my_data$sex)
my_data$sex <- droplevels(my_data$sex)

my_data$meovrwrk <- zap_missing(my_data$meovrwrk)
my_data$meovrwrk <- as_factor(my_data$meovrwrk)
my_data$meovrwrk <- droplevels(my_data$meovrwrk)
```

Checkpoint 03: Create and interpret confidence interval

A. Calculate the mean and 95% confidence interval for `lifenow`

```r
my_data |>
  dplyr::summarise(
    n = n(),
    mean_lifenow = mean(lifenow),
    sd_lifenow = sd(lifenow),
    lowCI_lifenow = CI(lifenow, ci = 0.95)['lower'],
    hiCI_lifenow  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 1 x 5
      n mean_lifenow sd_lifenow lowCI_lifenow hiCI_lifenow
  <int>        <dbl>      <dbl>         <dbl>        <dbl>
1  1448         7.73       1.65          7.64         7.81
```

Interpret your confidence interval (e.g., say in plain language what the CI is telling us).

The average life satisfaction score of 7.73 on a 0–10 scale, indicates generally high levels of satisfaction.

There is a 95% probability that the true population mean falls between 7.64 and 7.81. Or: We are 95% confident the true average life satisfaction score is between 7.64 and 7.81. OR: Based on the sample, there is 95% certainty that the population average life sanctification is between 7.64 and 7.81.

B. Calculate the mean and 99% confidence interval for `lifenow`

```r
my_data |>
  dplyr::summarise(
    n = n(),
    mean_lifenow = mean(lifenow),
    sd_lifenow = sd(lifenow),
    lowCI_lifenow = CI(lifenow, ci = 0.99)['lower'],
    hiCI_lifenow  = CI(lifenow, ci = 0.99)['upper']
    )
```

```
# A tibble: 1 x 5
      n mean_lifenow sd_lifenow lowCI_lifenow hiCI_lifenow
  <int>        <dbl>      <dbl>         <dbl>        <dbl>
1  1448         7.73       1.65          7.61         7.84
```

How does the 95% confidence interval differ from the 99% confidence interval in terms of width and certainty?

A 99% confidence interval of 7.61 to 7.83 life satisfaction points offers greater certainty that the true population mean is captured, but it is less precise than the narrower 95% interval of 7.64 to 7.81. The 99% interval is 0.05 life satisfaction points wider, reflecting that increased confidence comes at the cost of precision, as the estimate becomes less specific.

Checkpoint 04: Create, compare and interpret confidence intervals

A. Calculate the mean and 95% confidence intervals for `lifenow` by `college`

```r
my_data |>
  group_by(college) |>
  dplyr::summarise(
    n = n(),
    mean = mean(lifenow),
    sd = sd(lifenow),
    lowCI = CI(lifenow, ci = 0.95)['lower'],
    hiCI  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 2 x 6
  college                n  mean    sd lowCI  hiCI
  <chr>              <int> <dbl> <dbl> <dbl> <dbl>
1 College degree       638  7.93  1.55  7.81  8.05
2 No college degree    810  7.57  1.71  7.45  7.69
```

Do the confidence intervals for these two groups overlap? What does that imply about the statistical significance of their difference?

The non-overlapping 95% confidence intervals indicate a likely statistically significant difference in average life satisfaction, with college graduates reporting notably higher average life satisfaction (7.92) than those without a degree (7.56).

B. Calculate the mean and 95% confidence intervals for `lifenow` by `edu_category`

```
my_data |>
  group_by(edu_category) |>
  dplyr::summarise(
    n = n(),
    mean = mean(lifenow),
    sd = sd(lifenow),
    lowCI = CI(lifenow, ci = 0.95)['lower'],
    hiCI  = CI(lifenow, ci = 0.95)['upper']
    )
```

```
# A tibble: 3 x 6
  edu_category             n  mean    sd lowCI  hiCI
  <fct>                <int> <dbl> <dbl> <dbl> <dbl>
1 High school or less    419  7.56  1.61  7.41  7.72
2 Some college           391  7.57  1.81  7.39  7.75
3 College or more        638  7.93  1.55  7.81  8.05
```

Do the confidence intervals for these three groups overlap? What does that imply about the statistical significance of their difference?

The 95% confidence intervals for high school or less (7.40–7.71) and some college (7.39–7.75) overlap substantially, suggesting no statistically significant difference in average life satisfaction between those two groups. However, the interval for college or more group (7.80–8.04) does not overlap with either of the other groups, indicating a statistically significant and meaningfully higher level of life satisfaction among those with a college degree or more.

Checkpoint 05: Create, compare and interpret confidence intervals

Calculate the mean and 95% confidence intervals for `hurt_num` by `marstat` & `sex` (e.g, married women, married men, single women, single men, etc.)

```r
my_data |>
  group_by(marstat, as_factor(sex)) |>
  dplyr::summarise(
    n = n(),
    mean = mean(hurt_num),
    sd = sd(hurt_num),
    lowCI = CI(hurt_num, ci = 0.95)['lower'],
    hiCI  = CI(hurt_num, ci = 0.95)['upper']
    )
```

```
`summarise()` has grouped output by 'marstat'. You can override using the
`.groups` argument.
```

```
# A tibble: 6 x 7
# Groups:   marstat [3]
  marstat      `as_factor(sex)`     n  mean    sd lowCI  hiCI
  <fct>        <fct>            <int> <dbl> <dbl> <dbl> <dbl>
1 Single       male               240 0.496 0.501 0.432 0.560
2 Single       female             291 0.419 0.494 0.362 0.476
3 Married      male               350 0.563 0.497 0.511 0.615
4 Married      female             281 0.349 0.477 0.293 0.405
5 Prev married male               112 0.518 0.502 0.424 0.612
6 Prev married female             174 0.368 0.484 0.295 0.440
```

Assess whether gender differences within each marital status group are likely to be statistically meaningful.

Among singles and previously married individuals, the 95% confidence intervals for men and women overlap, indicating no statistically significant gender difference in beliefs about whether men hurt their families by overworking. In contrast, the married group shows non-overlapping intervals, suggesting a statistically significant difference between men and women on this issue.

Checkpoint 06: Write a null and research hypothesis

Write a null and research hypothesis for the association between the variables `lifenow` and `college`

H0: There is no difference in average life satisfaction scores between people with a college education and those without.

H1: People with a college education have higher average life satisfaction scores than those without.

Checkpoint 07: Produce and interpret a two-sample t-test

Use `t.test()` to produce a two-sided t.test of the variables `lifenow` and `college`

```
t.test(my_data$lifenow ~ my_data$college, alternative = "two.sided")
```

```

	Welch Two Sample t-test

data:  my_data$lifenow by my_data$college
t = 4.2025, df = 1418.3, p-value = 2.805e-05
alternative hypothesis: true difference in means between group College degree and group No co
95 percent confidence interval:
 0.1919569 0.5280400
sample estimates:
   mean in group College degree mean in group No college degree
                       7.927900                        7.567901
```

Explain what the test reveals about the statistical differences in average life satisfaction between the two groups. What do the results suggest about the null and research hypotheses?

The t-test result (t = 4.2025, p < .001) indicates a statistically significant difference in average life satisfaction between individuals with and without a college education. Therefore, we reject the null hypothesis.

Checkpoint 08: Create cross-tabs with chi-square tests

Use `tbl_cross()` and `add_p()` to create a cross-tab with a chi-square test for the variables `meovrwrk` and `college`

```
my_data |>
  tbl_cross(
    row = meovrwrk,
    col = college,
    percent = "column") |>
  add_p(source_note = TRUE)
```

Is there a statistically significant difference in agreement that men hurt the family when they focus on work too much by college degree?

	college		
	College degree	No college degree	Total
meovrwrk			
strongly agree	48 (7.5%)	75 (9.3%)	123 (8.5%)
agree	248 (39%)	287 (35%)	535 (37%)
neither agree nor disagree	178 (28%)	230 (28%)	408 (28%)
disagree	127 (20%)	181 (22%)	308 (21%)
strongly disagree	37 (5.8%)	37 (4.6%)	74 (5.1%)
Total	638 (100%)	810 (100%)	1,448 (100%)

Pearson's Chi-squared test, p=0.3

A chi-square test yielded a p-value of 0.3, indicating that the relationship between gender and agreement with whether men hurt the family when they overwork is not statistically significant. Therefore, we fail to reject the null hypothesis that college degree and agreement are independent.

Checkpoint 09: Calculate and interpret a chi-square test

Use `tbl_cross()` and `add_p()` to create a cross-tab with a chi-square test for the variables `meovrwrk` and `sex`

```
my_data |>
  tbl_cross(
    row = meovrwrk,
    col = sex,
    percent = "column") |>
  add_p(source_note = TRUE)
```

Is there a statistically significant gender difference in agreement that men hurt the family when they focus on work too much?

A chi-square test yielded a p-value less than .001, so we reject the null hypothesis. There appears to be a statistically significant relationship between gender and agreement with the statement that men hurt the family when they overwork.

Checkpoint 10: Produce and interpret a Pearson's Correlation Coefficient (R)

Use `cor.test()` to test the correlation between `lifenow` & `educ`

	sex		
	male	female	Total
meovrwrk			
strongly agree	76 (11%)	47 (6.3%)	123 (8.5%)
agree	298 (42%)	237 (32%)	535 (37%)
neither agree nor disagree	185 (26%)	223 (30%)	408 (28%)
disagree	124 (18%)	184 (25%)	308 (21%)
strongly disagree	19 (2.7%)	55 (7.4%)	74 (5.1%)
Total	702 (100%)	746 (100%)	1,448 (100%)

Pearson's Chi-squared test, p<0.001

```r
cor.test(
  ~ lifenow + educ,
  data = my_data,
  method = "pearson",
  na.action = na.omit
)
```

```

	Pearson's product-moment correlation

data:  lifenow and educ
t = 4.9847, df = 1446, p-value = 6.956e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.07898701 0.18028011
sample estimates:
      cor
0.1299727
```

Describe the strength and direction of any correlation between education and life state.

The Pearson correlation test reveals a statistically significant ($p < .001$) but weak positive relationship ($r = 0.12$) between education and life satisfaction, suggesting that higher education levels are modestly associated with greater life satisfaction.

IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

- Choose an interval-ratio (e.g, continuous or proportion) variable that you wish to consider as a **dependent variable** (DV).
- Create a summary table for this DV, including the total number of observations, mean, median, and sd.
- Pick or create a dichotomous variable (e.g., gender, two racial categories; two age categories; two countries; two years; etc.) in your dataset and create confidence intervals for each group for your DV. Compare and interpret the confidence intervals.
- Write a null and research hypothesis using these two variables.
- Conduct a two sample t-test (using `t.test()`) for your DV and dichotomous variable. Interpret your results. How do they relate to your hypotheses?
- Choose a second interval-ratio variable that you think may be associated with your DV.
- Use `cor.test()` to formally test for an association between the two continuous variables. Interpret your results.